

VARUNKUMAR SONAWANE

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Education

Indiana University Bloomington, IN, USA.

Aug 2024 – May 2026

Master of Science in Computer Science

CGPA: 3.83/4

Certifications

AWS Certified Developer – Associate | Microsoft Azure AI Fundamentals

Experience

Data Engineer Intern

May 2025 - Aug 2025

The Commons XR

San Diego, CA

- Eliminated manual data handoffs by engineering **batch pipelines** to integrate and migrate **5M+ records/week** from **OLTP PostgreSQL to OLAP BigQuery using Datastream** ensuring **data quality and validation**.
- Strengthened data-driven decision making by communicating insights through the development and maintenance of **10+** interactive **Plotly and Dash** dashboards, supporting ad-hoc analysis for XR session analytics.
- Automated post-session behavioral analytics for **2 XR sessions per week** by replacing manual analysis with an **LLM-powered RAG pipeline** in Vertex AI that generated session-level insights and dashboards.
- Improved analytics readiness by delivering analytics-ready datasets across bronze, silver, and gold layers using the **Medallion Architecture** in BigQuery as part of a cloud data warehouse.

Graduate Teaching Assistant

Dec 2025 - Present

Indiana University

Bloomington, IN

- Supported applied data engineering workflows for **30+ graduate students** by guiding hands-on labs in **Scala and Apache Spark**, covering distributed processing, DataFrames, transformations, and aggregations.
- Advanced analytics and **unstructured data** analysis by supporting **10+ Tableau** labs and NLP assignments, including dashboard design, text extraction, cleaning, and analysis using NLTK, spaCy, and Python pipelines.
- Guided database fundamentals and data preparation for 120+ undergraduate students by reinforcing relational **data modeling**, SQL querying, schema design, normalization, and reducing preprocessing errors by **~20%** across time series, graph data, and Machine Learning oriented transformations.

Software Engineering Intern

July 2023 - Jan 2024

CodeClause

Pune, India

- Improved code quality in a scalable service architecture by stabilizing backend services, adding unit tests, and fixing debugging and error handling issues across **5+ REST API** modules within a **CI/CD-driven DevOps workflow**.
- Enhanced system reliability and observability by applying **system design** principles and implementing structured **logging** and **monitoring** across multiple services, reducing recurring runtime failures.

Projects

ShieldScrapper – Cloud-Based Web Scraping & Analytics Pipeline | *Glue, Athena SQL, DynamoDB, Docker*

- Automated daily data ingestion of **10K+ JSON** records by running containerized Scrapy spiders on ECS Fargate, scheduled via EventBridge, and storing raw data in Amazon S3.
- Delivered daily refreshed analytics across multiple leagues by executing **AWS Glue ETL jobs**, querying via Athena SQL, loading results into **DynamoDB**, and publishing QuickSight dashboards, with Lambda + SNS alerts.

Big Data Analytics with AWS and PySpark | *Pyspark, Lambda, S3, EC2, SNS, Sagemaker, Power BI*

- Built an end-to-end big data pipeline processing **100K+ records** by ingesting data from **Amazon S3**, handling schema and **file-level metadata** and performing distributed transformations using **PySpark on EC2**.
- Enabled **KPI-driven analytics** by orchestrating AWS Lambda to trigger **SageMaker AutoML** on processed S3 data and publishing model outputs to **Power BI** dashboards, with **SNS** notifications for pipeline execution.

Weather Data ETL Pipeline with Airflow | *Airflow, Snowflake, Astronomer*

- Orchestrated **Airflow DAGs** to extract weather data from the OpenWeather API for 10+ locations.
- Scheduled daily ETL workflows to transform and load **time-series** weather data into **Snowflake**.

Technical Skills

Programming & Scripting: Python, SQL, Bash.

Databases & Cloud: PostgreSQL, MySQL, NoSQL, MongoDB, Cassandra, Neo4j, AWS, Azure, GCP, Snowflake.

ML & Analytics: Scikit-learn, TensorFlow, Pandas, NumPy, LangChain, Prompt Engineering, Feature Engineering.

Big Data & ETL: Airflow, PySpark, Hadoop, Databricks, DBT, Iceberg, Kinesis, Redshift, AWS Glue, Kafka, Datadog.

BI Tools & Data Visualization: Power BI, Tableau, Excel, Git, Linux CLI, VS Code, System Design, Agile, Statistics.